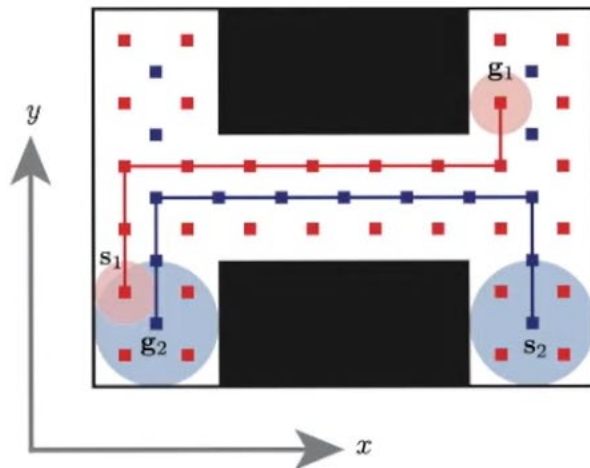


Coordination of multiple robots

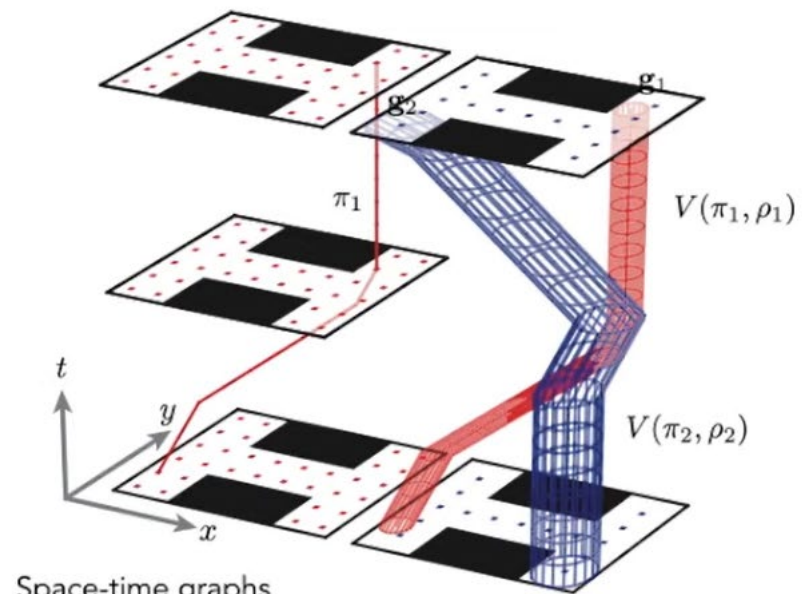
- n robots $R_1, \dots, R_n$, with configuration spaces $C_1, \dots, C_n$, sharing the same workspace
- **Problem:** Plan coordinated motion so that each robot achieves its own goal configuration.
- **Centralized planning:** Plan the coordinated motion in $C_1 \times C_2 \times \dots \times C_n$ (but very high dimensional space)
- **Decoupled planning:** Plan the motion of each robot ignoring the other robots; then coordinate their motions so that no two robots collide
 - **Prioritized planning:** Plan the motion of one robot ignoring the other robots; then plan the trajectory of a second robot in its configuration $\times$ time space treating the first robot as a moving obstacle; then plan the trajectory of a third robot ...

Planning with Moving Obstacles

- Dealing with moving obstacles require planning a **trajectory** τ , i.e., a path indexed by time
- τ is conveniently represented in **configuration x time space (CT-space)** by a continuous curve whose tangent always projects positively along the time axis
- Obstacles map as forbidden regions in CT-space. Constraints on velocity constrain tangents to τ . Constraints on acceleration constrain curvature of τ .



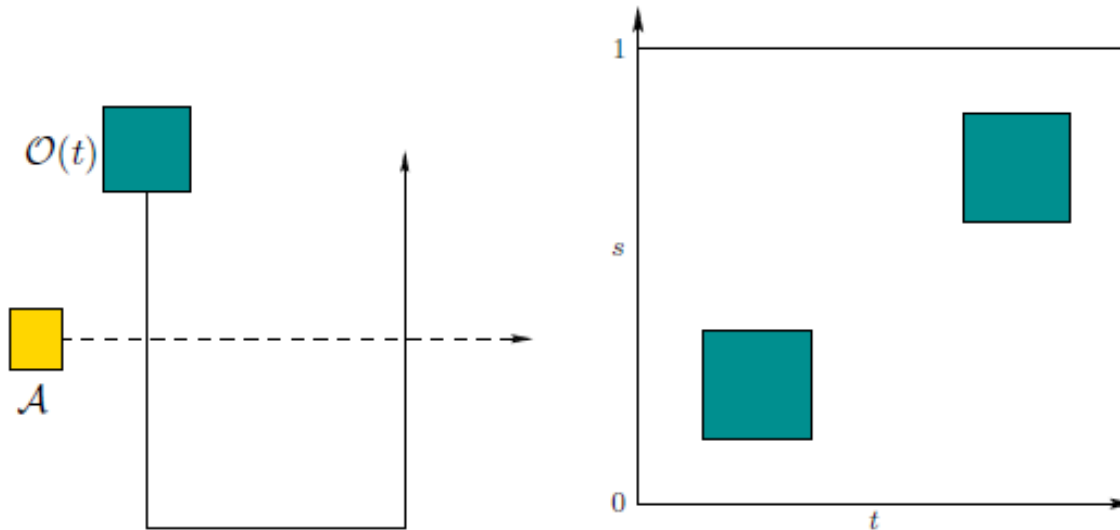
Ideal trajectories for 2 robots



Space-time graphs

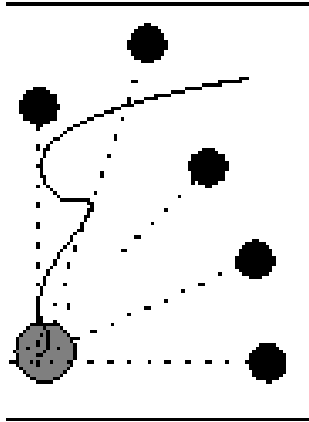
Decoupled Planning

- Pure velocity tuning:
 - (1) Separately plan a path of each robot to avoid collision with (static) obstacles
 - (2) Compute the relative velocities of the robots to avoid inter-robot collision (e.g., coordination diagram)

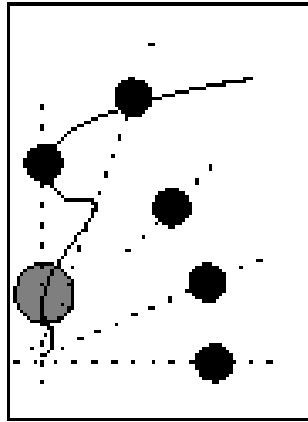


Example

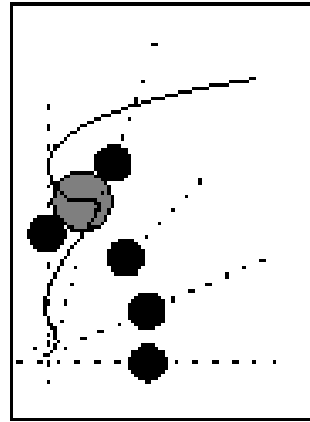
$T = 0.0 \text{ secs}$



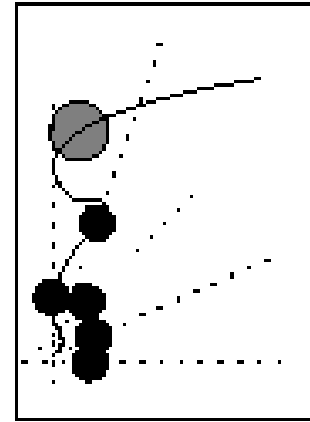
$T = 9.0 \text{ secs}$



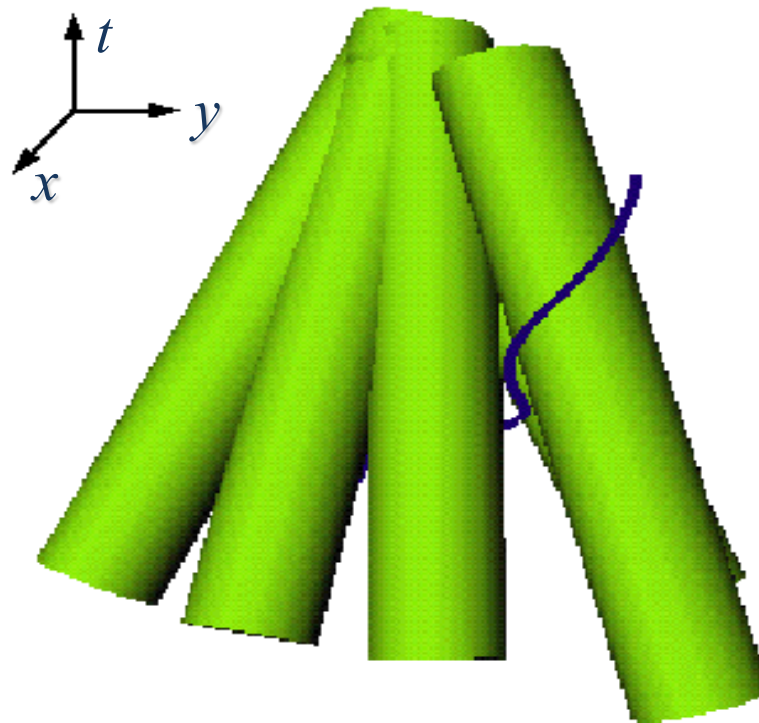
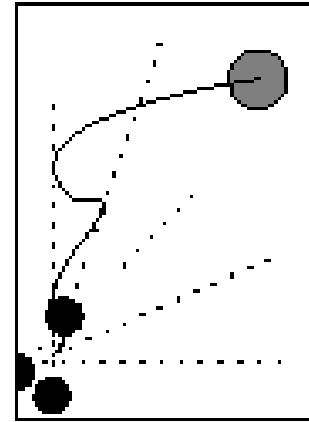
$T = 20.0 \text{ secs}$



$T = 30.0 \text{ secs}$

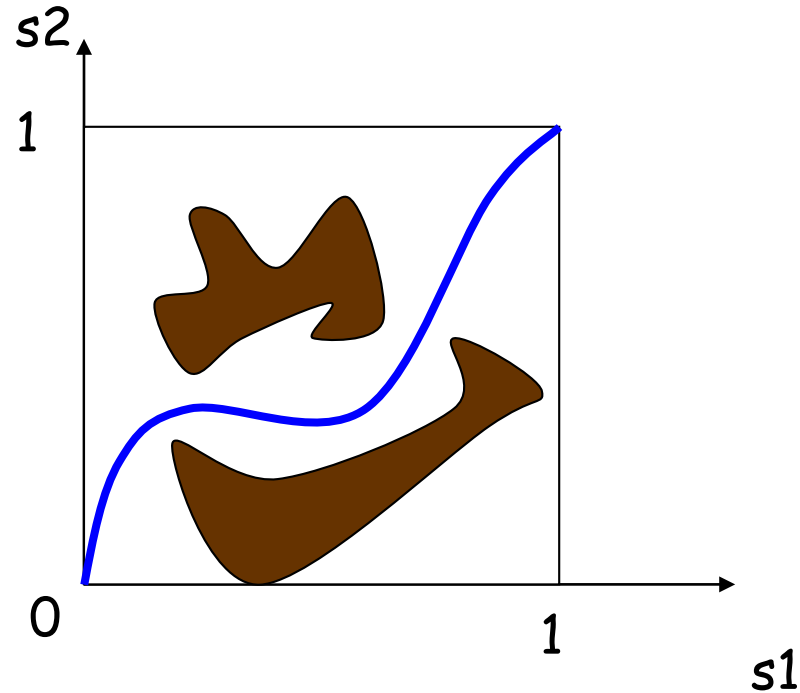


$T = 45.5 \text{ secs}$



Coordination Space

- 2 robots R1 and R2
- 2 paths: $t_i : s_i \in [0,1] \rightarrow C_i$ ($i=1,2$)
- 2-D coordination space



- Generalize to n robots $\rightarrow n$ -D coordination space
- Coordinate paths of R_1 and R_2 in a 2-D coordination diagram ($\rightarrow$ path of “ R_1 - R_2 ”), then coordinate paths of R_1 - R_2 and R_3 in 2-D coordination diagram, etc...

Decoupled Planning

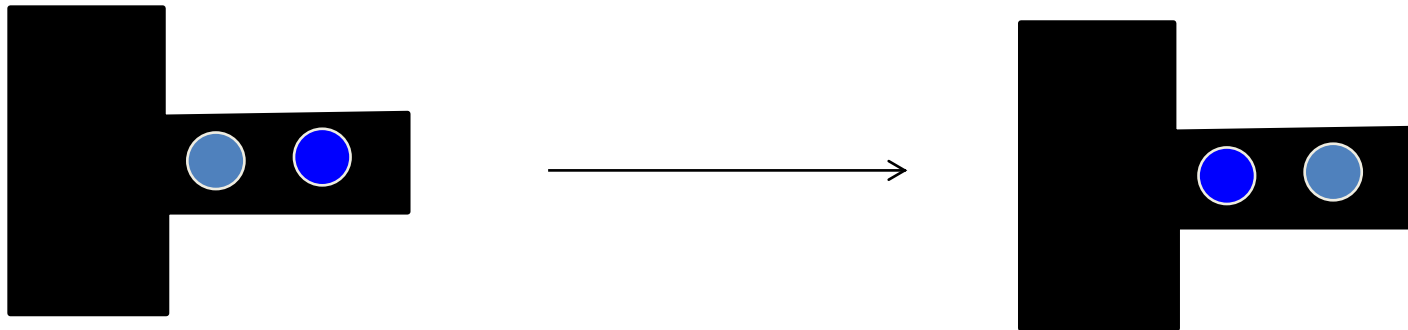
- Pure velocity tuning:
- Robot prioritization:
 - Plan path of a first robot in its C-space
 - Iterate: Plan trajectory of i -th robot in its CT-space assuming that robot-1,...,robot-(i -1) are obstacles moving at some velocities

Centralized Planning

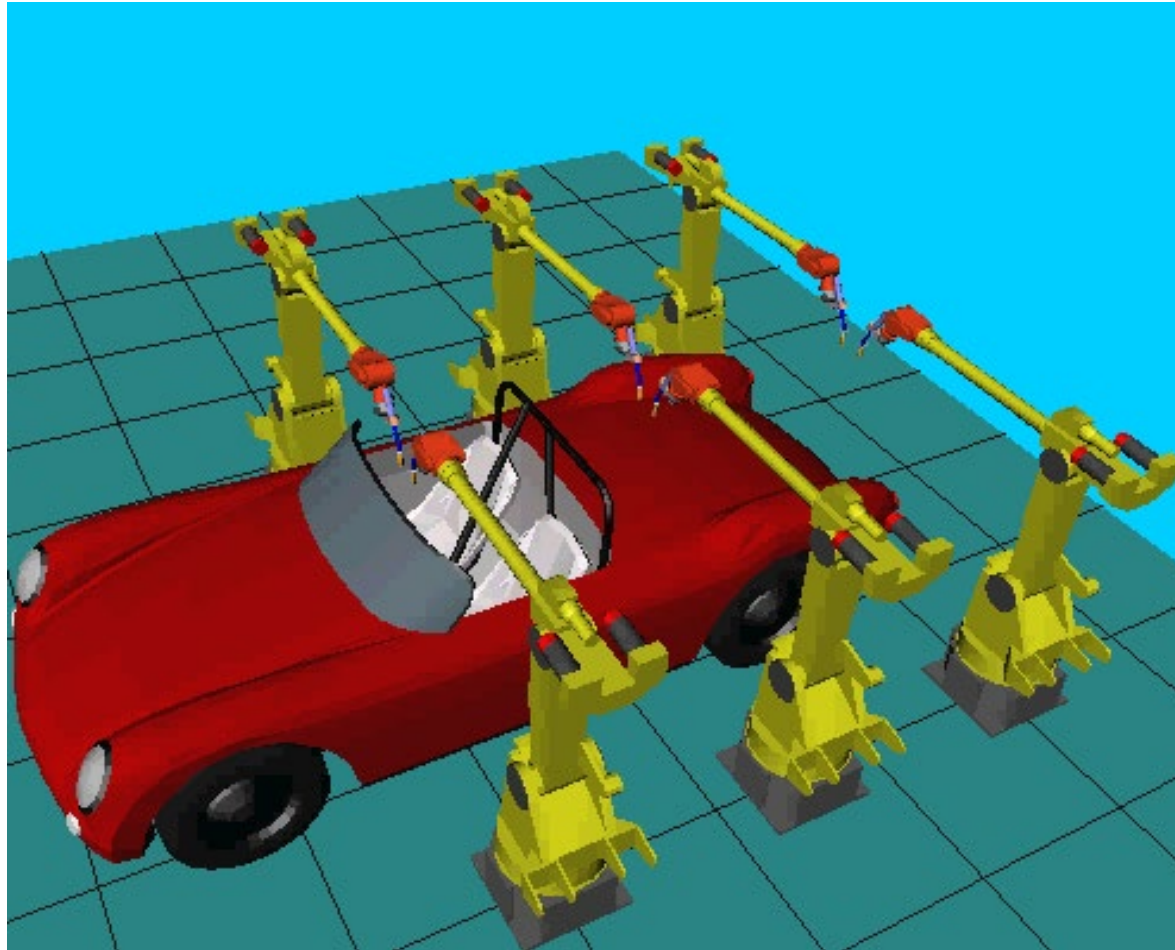
- Plan collision-free path τ in composite configuration $C_1 \times C_2 \times \dots \times C_p$ space of the p robots
- Forbidden regions in composite C-space are all configurations where either a robot collide with an obstacle or two robots collide with each other
- The projection of τ into C_i is the path of the i -th robot

Pros and Cons

- Assume p robots with n degrees of freedom each.
- Worst-case complexity of centralized planning is $\sim \exp(np)$
- Worst-case complexity of decoupled planning is $\sim p \exp(n) \ll \exp(np)$
- But decoupled planning is incomplete



Multi-Robot Example



- Centralized planning
- 36 dofs